

■ Customer Segmentation Analysis

Interactive Python Data Analysis Notebook

Important note

This PDF is an interactive **notebook-style guide**: it contains the project workflow, code, outputs, and places to experiment with the segmentation rules. A PDF itself cannot execute Python or Streamlit code; use the supplied Jupyter Notebook and app.py for live interaction.

1. Project Overview

Analyze 20 customers using total spending and purchase frequency, then classify them into High Value, Regular, and Low Value customers using simple Python rules. No machine-learning algorithm is used.

2. Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

3. Create Customer Dataset

```
data = {
    "CustomerID": range(1, 21),
    "CustomerName": ["Rahul", "Priya", "Amit", "Sneha", "Rohan", "Anjali", "Vikash", "Neha", "Arjun", "Pooja",
    "Karan", "Riya", "Aditya", "Nisha", "Sahil", "Kavya", "Manish", "Simran", "Ravi", "Isha"],
    "Total_Spending": [85000, 72000, 15000, 22000, 95000, 18000, 65000, 42000, 12000, 55000,
    78000, 25000, 90000, 35000, 14000, 68000, 30000, 82000, 20000, 48000],
    "Purchase_Frequency": [18, 15, 4, 5, 20, 3, 14, 10, 2, 12, 17, 6, 19, 8, 3, 13, 7, 16, 5, 11]
}
df = pd.DataFrame(data)
df
```

4. Data Check

```
print("Number of Customers:", len(df))
print("\nMissing Values:")
print(df.isnull().sum())
print("\nBasic Statistics:")
print(df[["Total_Spending", "Purchase_Frequency"]].describe())
```

5. Rule-Based Segmentation

```
def segment_customer(row):
    spending = row["Total_Spending"]
    frequency = row["Purchase_Frequency"]

    if spending >= 60000 and frequency >= 12:
        return "High Value Customer"
    elif spending >= 30000 and frequency >= 7:
        return "Regular Customer"
    else:
        return "Low Value Customer"

df["Segment"] = df.apply(segment_customer, axis=1)
df
```

6. Segment Summary

```
segment_summary = df.groupby("Segment").agg(
    Customers=("CustomerID", "count"),
    Average_Spending=("Total_Spending", "mean"),
    Average_Purchases=("Purchase_Frequency", "mean")
).round(2)

segment_summary
```

7. Visualization

```
plt.figure(figsize=(8, 5))

for segment in df["Segment"].unique():
    segment_data = df[df["Segment"] == segment]
    plt.scatter(
        segment_data["Total_Spending"],
        segment_data["Purchase_Frequency"],
        label=segment,
        s=100
    )

plt.title("Customer Segmentation")
plt.xlabel("Total Spending")
plt.ylabel("Purchase Frequency")
plt.legend()
plt.show()
```

8. Segment Distribution

```
segment_counts = df["Segment"].value_counts()

plt.figure(figsize=(7, 5))
segment_counts.plot(kind="bar")
plt.title("Number of Customers by Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Number of Customers")
plt.xticks(rotation=20)
plt.show()
```

9. Explore a Segment

```
selected_segment = "High Value Customer"

selected_customers = df[df["Segment"] == selected_segment]
selected_customers
```

10. Export Results

```
df.to_csv("customer_segmentation_result.csv", index=False)
print("File saved successfully!")
```

11. Interactive Streamlit App

```
# Run from Jupyter:
!streamlit run app.py

# Or from Command Prompt / Terminal:
# streamlit run app.py
```

■ Project Results & Interactive Checklist

Use this page as a compact project review.

Item	What to check
Customer records	20 customers are included in the notebook dataset.
Missing values	The notebook checks every column using <code>df.isnull().sum()</code> .
Segmentation	Customers are classified using spending and purchase-frequency rules.
Visualization	Scatter plot shows spending against purchase frequency.
Distribution	Bar chart shows the number of customers in each segment.
Interactive app	<code>app.py</code> provides threshold controls, summaries, charts, segment exploration and CSV download.

■ Final Project Files

Customer Segmentation Analysis.ipynb — complete analysis notebook

app.py — interactive Streamlit application

requirements.txt — required Python libraries

README.md — GitHub project documentation

How to use the interactive version

Open the notebook in Jupyter to execute and modify cells. For the dashboard, run **streamlit run app.py**. The sidebar controls let you change the high-spending and high-frequency thresholds and immediately inspect the resulting customer segments.